
Supplementary information

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representations in diverse visual scenes**

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Supplementary Information

Generation of stable heading representations in diverse visual scenes

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A model of E-PG neurons

The ring attractor dynamics in the (E-PG) compass neuron population result from the presence of local excitation and long-range inhibitory interactions between compass neurons (See ^{32,33} for detailed mathematical analyses). The firing rate dynamics of the compass neurons are described by

$$\tau \partial_t f_n = -f_n + \left[\alpha f_n + D(f_{n+1} + f_{n-1}) - \beta \sum_{m=0}^{N-1} f_m + 1 - \frac{v(f_{n+1} - f_{n-1})}{2} + I_n \right]_+,$$

where $[x]_+ = x\Theta(x)$, Θ is the Heaviside step function, f_n represents the activity of a compass neuron n , τ represents a decay constant set to 0.05 seconds, α, β and D denote respectively the strength of the recurrent interaction of a neuron to itself, uniform inhibition across all compass neurons, and excitation from neighboring neurons. The constant term, 1, is a uniform excitatory drive that generates activity in the system.

The domain of parameter values where the network dynamics exhibits a compass-neuron bump of activity has been analytically derived in our previous work³² and any set of values in this domain can be used to perform a stable simulation of ring attractor dynamics. We chose $\alpha = -7.76, \beta = 1.96, D = 5.19$ for a model with 32 compass neurons.

We modeled an internally generated angular turning signal as a single parameter, v , which modulates the interaction between neighboring neurons to shift the compass-neuron bump in one direction or another. For simplicity, v was determined by simulation for a given set of

parameters. Specifically, we estimated the v values required to induce a compass-neuron bump rotation with angular velocities of $0.5 \pi/sec$, π/sec , and $2\pi/sec$. In simulation, we interpolated v for other desired angular velocities.

A model of visual input neurons

The external input to a compass neuron n is described by I_n . Specifically,

$$I_n = -\mathbf{w}_n \mathbf{g} + s_n = -\sum_m w_{n,m} g_m + s_n$$

or in vector form

$$\mathbf{I} = -\mathbf{W} \mathbf{g} + \mathbf{s},$$

where $\mathbf{I} = [I_1, I_2, \dots, I_{n_{max}}]^T$, $\mathbf{w}_n = [w_{n,1}, w_{n,2}, \dots, w_{n,m_{max}}]$, $\mathbf{g} = [g_1, g_2, \dots, g_{m_{max}}]^T$, $\mathbf{s} = [s_1, s_2, \dots, s_{n_{max}}]^T$, $n_{max} = 32$, $m_{max} = 32$, g_m represents the activity of a ring neuron m , $w_{n,m}$ represents the strength of an inhibitory synaptic weight from a ring neuron m to a compass neuron n , and $\mathbf{W}$ represents the entire synaptic weight matrix. In our simulations the weights are within the range $0 \leq w_{n,m} \leq w_{max} = 0.33$. s_n represents other external inputs, such as current injections by optogenetic stimulation to a compass neurons n .

Modeling one-dimensional (1D) scenes

The population activity of ring neurons represents visual scenes. In our model, ring neurons have a homogeneous receptive field (RF) structure with RF locations evenly distributed along the azimuth. The activity of each visual neuron represents the light intensity of a visual scene at the angular position where the RF is located.

We used three types of 1D visual scenes in this work.

- (1) A narrow von Mises function with a concentration parameter $\kappa = 15$ to represent a single stripe.
- (2) Two narrow von Mises functions to represent two stripes 180 degrees apart.
- (3) Mixture of four von Mises functions with random angular positions and random κ to simulate a complex scene.

The minimum of the ring neuron activity was set to zero. The maximal activity was chosen, using a single narrow von Mises function, so that the visual input is weak enough not to dominate the ring attractor dynamics in case the synaptic weight matrix is random but strong enough to bias the compass-neuron bump position after the synaptic weight matrix is learned through each Hebbian rule. Once set, the same maximum activity level was used to scale all other visual stimuli. During simulation we always added noise to visual input neurons. Specifically, samples from a uniform distribution with a range of 50% of the maximum input activity was added to the visual input neurons, then box-car averaging over five time points (25 ms for simulations with temporal resolution of 5 ms) was applied.

Modeling two-dimensional (2D) scenes

For Extended Data Fig. 6, 64 visual ring neurons were arranged into 32 pixel width by 2 pixel height scenes. For the simulation of memory interference (Extended Data Fig. 6), a narrow 1D von Mises function (as in the 1D scene) was placed either at the top row or at the bottom row. Conceptually, each row represents a population of input neurons that could encode distinct features, not just different RF locations. For example, the top row may represent ring neurons encoding the blue hue, whereas the bottom row may represent those encoding the green hue. So, a blue scene may only activate neurons at the top row, and a green scene may only activate the bottom row. Note that this is a fictive example. The actual feature space of the visual ring neurons may have different and more diverse attributes¹³⁻¹⁶. During simulation, we always added noise to visual inputs neurons.

For Extended Data Fig. 5, 256 visual ring neurons were arranged into 32 pixel width by 8 pixel height scenes. Two artificial scenes shown in Extended Data Fig. 5 were converted to 8x32 pixel scenes. To account for the receptive field size of the ring neurons^{13,39}, we applied 2D gaussian filters ($\sigma = 15^\circ$).

Modeling angular velocity and corresponding scene orientation

To simulate realistic flight behavior, we took 400 s of tethered flight data to infer an angular turning signal v , with which a simulation of compass neurons in the absence of visual input would generate a temporal compass-neuron bump position profile identical to the observed real scene orientation profile. The temporal profile of the simulated visual scene orientation was also made to match the same profile. For longer simulations, the same 400 s segment was repeated. During simulation, we always added noise to v . Specifically, samples from a uniform distribution with a range of 20% of the maximum velocity for each simulation was added to the angular velocity profile, then box-car averaging over five time points was applied.

Models of Hebbian learning rules

In principle, many learning rules could account for the majority of physiological data in this study. We considered two classes of Hebbian learning rules: (i) Post-synaptically gated, where plasticity only occurs when a post-synaptic compass neuron is active, and (ii) Pre-synaptically gated, where plasticity only occurs when a pre-synaptic ring neuron is active. There are many possible implementations of these two classes of learning rules, which would produce qualitatively similar results within classes. In our simulations, we considered the following specific implementations:

Post-synaptically gated rule

$\partial_t w_{n,m} = \eta [f_n - f_{th}]_+ \{w_{max}(1 - g_m/g_0) - w_{n,m}\}$, $\eta = \varepsilon v^2$, where $g_0 = 0.33$ is a scaling factor, $w_{max} = 0.33$, and η is the learning rate, which depends on the angular velocity v to compensate for uneven sampling of scene orientations over time in case, e.g., a fly fixates on a single orientation for an extended period of time. $\varepsilon = 0.5$ is a scalar, and $f_{th} \geq 0$ was set to

zero. This Hebbian rule is mathematically similar to Kohonen's Self Organizing Map^{62,63} and is conceptually similar to other approaches^{19,20,22,64,65}. Most simulation figures here and all previously specified parameter values in Supplementary Information correspond to this rule. After learning, the synaptic weight profile from all ring neurons to a single compass neuron matches the activity profile of the ring neurons that provide input to the compass neuron.

Pre-synaptically gated rule

$\partial_t w_{n,m} = \eta [g_m - g_{th}]_+ \{w_{max}(1 - f_n/f_0) - w_{n,m}\}$, $\eta = \varepsilon v^2$, where $f_0 = 0.08$ is a scaling factor, $w_{max} = 0.33$, and $g_{th} \geq 0$ was set to 0.1, which is approximately a third of w_{max} . g_{th} may be dynamically adjusted using the past few seconds' average of neural activity, although it has not been extensively tested.

In all models, the plasticity requires both potentiation and depression of the synaptic weight.

Simulation of the entire model

Simulation results in this study began with a random synaptic weight matrix, but simulations beginning with a weight matrix with constant w_{max} or 0 behaved identically. In all simulations, the compass population activity began with a random initial activity profile, which quickly settled into a single compass-neuron bump during the initial seconds of simulation. The temporal resolution of the model used to generate figures was 5 ms. For animation (Supplementary Video 3), the temporal resolution was 50 ms, the learning rate scalar was increased to $\varepsilon = 4$ to speed up the simulation, and noise was not added to ring neuron activity for visual clarity. Temporal resolution of simulations in both Extended Data Fig. 5 and 6 was 50 ms. For Extended Data Fig. 5, $\varepsilon = 1.5$ (post-synaptic rule) and $\varepsilon = 3$ (pre-synaptic rule) were used to increase simulation speed, and 75 seconds of the flight data was simulated to reduce the simulation time. The maximum input was also adjusted to a half for the post-synaptic rule and to a third for the pre-synaptic rule to counter the excessive inhibition from the increased number of inhibitory visual neurons.

Memory capacity of different learning rules

Simple artificial scenes with identical local features at the same elevation and azimuth but in different arrangements produced deterministic offset switches across flies (Extended Data Fig. 5a-d). The difference between offsets for the two scenes was exactly as predicted from a 2D cross-correlation of ring-neuron filtered versions of these scenes^{13,39} (Extended Data Fig. 5e-g) and is well-captured by our simulations (Extended Data Fig. 5g-k). Such deterministic offset switches between scenes make it difficult to determine whether or not the compass circuit can generate and store independent heading representations for different scenes. A capacity for storing multiple scene offsets is a likely pre-requisite for snapshot-based navigation^{30,31,40,41,66}, which requires retrieving and generating scene-specific directional actions.

These deterministic changes in offset between scenes may be a consequence of the simplicity of the artificial scenes. In contrast to these simple scenes, many natural scenes (Fig. 5a) are composed of a diversity of visual features at different elevations. If these features excite non-overlapping ring neuron populations^{13,14,35,39}, the compass system might be able to generate and store unique heading representations for each different natural scene. Indeed, for natural scenes

that produced a stable heading representation (open-space and forest scenes in Fig. 5a), the compass-neuron bump maintained the same pinning offset across multiple encounters of the same natural scene, even when these encounters were separated in time by an encounter with a different natural scene (Fig. 1h-j, Extended Data Fig. 6e). Importantly, these offsets were retrieved successfully despite the fact that changes in offset between the two natural scenes varied significantly across flies (Extended Data Fig. 6e; compare with Extended Data Fig. 5d). These observations indicate a capacity for storing multiple scenes without interference.

In the model, the capacity to support multiple maps depends on how the synaptic plasticity rule is gated by activity. The rule used in Fig. 2 – 4 and Extended Data Fig. 2 – 4 is post-synaptically gated, meaning that plasticity requires post-synaptic activity. Such rules modify synapses if the compass neurons are active regardless of ring neuron activity, thereby resetting synaptic weights from quiet ring neurons to active compass neurons. This destroys any information that those synapses might carry about previous visual maps, limiting storage capacity (Extended Data Fig. 6a, b, d, g). On the other hand, with pre-synaptically gated mechanisms, synapses are modified only if ring neurons are active, preserving synapses from quiet ring neurons. Models with pre-synaptic gating can account for the previously-described physiological data, but they have a significantly higher storage capacity. Specifically, in the extreme case when two scenes activate mutually-exclusive ring neuron populations, these rules allow the bump offsets of both scenes to be stored without any interference (Extended Data Fig. 6a, b, c, f). These modeling results suggest that if the overlap between the ring neuron populations activated by different natural scenes is small enough, pre-synaptic plasticity could explain the observation that bump offsets differ across scenes but are maintained for different encounters of the same scene (Extended Data Fig. 6e, c and f; see Supplementary Information for further discussion). Note, however, that even with post-synaptically gated rules, if pre-existing weights are not completely erased, the offset may still be recovered.

Bump offset changes under different plasticity rules

In an environment with two oppositely-positioned vertical stripes (Extended Data Fig. 3), simulations relying on either plasticity rule would suggest that weights from ring neurons to compass neurons should be modified as simulated in Extended Data Fig. 3a-c. However, in our experiments, the bump was largely confined to half the EB in most flies, a network asymmetry that we cannot yet explain. In this scenario, the two models predict different synaptic weight profiles after two-stripe exposure. For instance, let's assume that the compass-neuron bump dynamics were confined to the upper half of the weight matrix shown in Extended Data Fig. 3a. With the post-synaptically gated rule, the lower half of the weight matrix would not be modified. Thus, the lower right corner would be free of a dark band after exposure to the two-stripe environment. On the other hand, with a pre-synaptically gated rule, the lower half of the weight matrix would be erased.

The plasticity rules would also differ in their predictions of global extrapolation of offset changes after a limited experience of scene orientations as in Extended Data Fig. 4. In Extended Data Fig. 4a, for instance, the simulation used a post-synaptically gated rule, which erased the upper right corner of the synaptic weight matrix. In the same condition, a pre-synaptically gated rule would erase the lower left corner instead. However, this difference would produce qualitatively similar

neural dynamics. Testing such subtle differences in predictions will require new tools for in vivo, real-time monitoring of synaptic weights between ring neurons and compass neurons.

Other aspects of plasticity rules to explore in the future

Several features of the plasticity rules we explored would benefit from further investigation:

- 1) We have implemented learning rules with a nonlinear dependence on post- and/or pre-synaptic activities. For post-synaptically gated rules, synapses are potentiated or depressed depending on whether or not pre-synaptic activity exceeds a threshold, and vice versa for pre-synaptically gated rules. Research into other learning rules, such as BCM⁶⁷, has suggested molecular mechanisms that could implement such a threshold and dynamically adjust its properties in neurons⁶⁸. The observed plasticity in our study, if confirmed to occur at synapses between ring neurons and compass neurons, could offer an accessible system to investigate the biophysics of these synaptic nonlinearities in the *Drosophila* brain.
- 2) Synaptic potentiation/depression dynamics: We assumed the same time constants for both potentiation and depression, but the dynamics of these two processes might differ.
- 3) The plasticity rules that we considered fall into two extremes, being gated either pre- or post-synaptically. However, the actual rule could be a combination of both.
- 4) The learning rate of these rules was assumed to be a function of the fly's turning velocity, which was introduced to compensate for biased sampling of angular positions of the scene. Neuromodulatory systems in *Drosophila* are well known to be differentially activated by behavioral context and arousal^{36,57,69-72}. Similar mechanisms, if they exist in the navigation system, may contribute to the adaptive learning rate postulated in the proposed learning rules.

Code

MATLAB code for all the simulations is freely available at:

http://research.janelia.org/jayaraman/Kim_etal_Nature2019_Downloads/

To run the simulations, execute “main_sim.m”. Simulations can be configured in “main_config.m”. Learning rules are implemented in “RingAttractorODESolver.m”. Other files define simulation parameters, test integrity of the ring attractor model, and display the results.

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